Week 6 Material - Defining and Estimating Integrals as Areas

24. [1] Oil is leaking out of a yacht damaged at sea. The damage to the yacht is worsening, as indicated by the increased leakage rate each hour, recorded in the following table:

Time (h)	0	3	6	9	12
Oil Leakage Rate (litre/h)	114	159	235	295	404

Using these data with a pair of Riemann sums, find reasonable upper and lower estimates of the total quantity of oil that has leaked out over the 12 hours.

25. Find the **exact** value of the integral $\int_{-3}^{0} (1 + \sqrt{9 - x^2})\, dx$ by interpreting it in terms of areas. Hint: a sketch will be essential.

26. [2] A car's velocity at the beginning of a trip was tracked for one hour, with the following results.

t (hrs)	0	0.2	0.4	0.6	0.8	1
$v(t)$ (km/hr)	10	30	30	70	90	90

(a) Use a **right sum** to estimate the distance travelled by the car between $t = 0$ and $t = 1$ hour. Give units in your answer.

(b) A car's mileage or gas consumption depends on its speed. For the same car as in part (a), the table below shows the relationship between the speed the car is travelling (v) and the resulting gas consumption (reported in liters required to travel 100 km).

v (km/hr)	10	30	50	70	90	110
Efficiency (liters per 100 km)	29	15	10	7	8	9

Using a right-sum approach for each time interval, use this efficiency information and the trip information from part (a) estimate the amount of gas consumed by the car between $t = 0$ and $t = 1$ hour. Give units in your answer.

27. Consider the integral representing the area under the graph of $f(x) = \cos(x)$ from $x = 0$ to $x = \pi/2$.

(a) Use the trapezoidal rule with four intervals to estimate this area.

(b) Sketch the graph and your trapezoidal estimate of the area. Is your estimate an underestimate or an overestimate of the exact integral value?

[1] Source: previous APSC 171 Test 2
[2] Source: previous APSC 171 Test 2